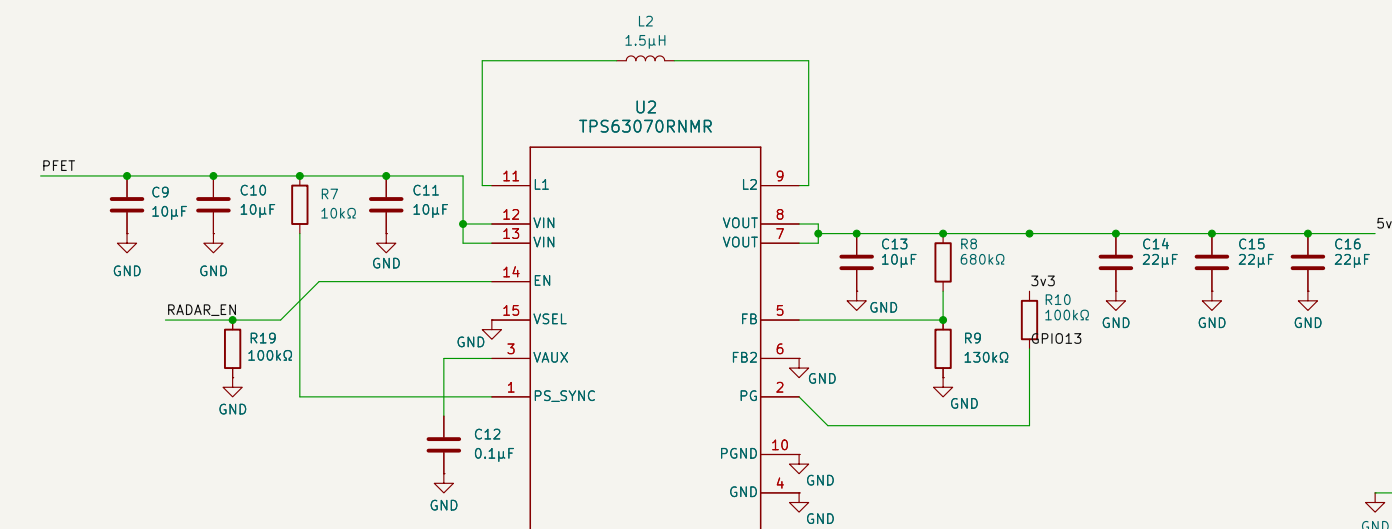
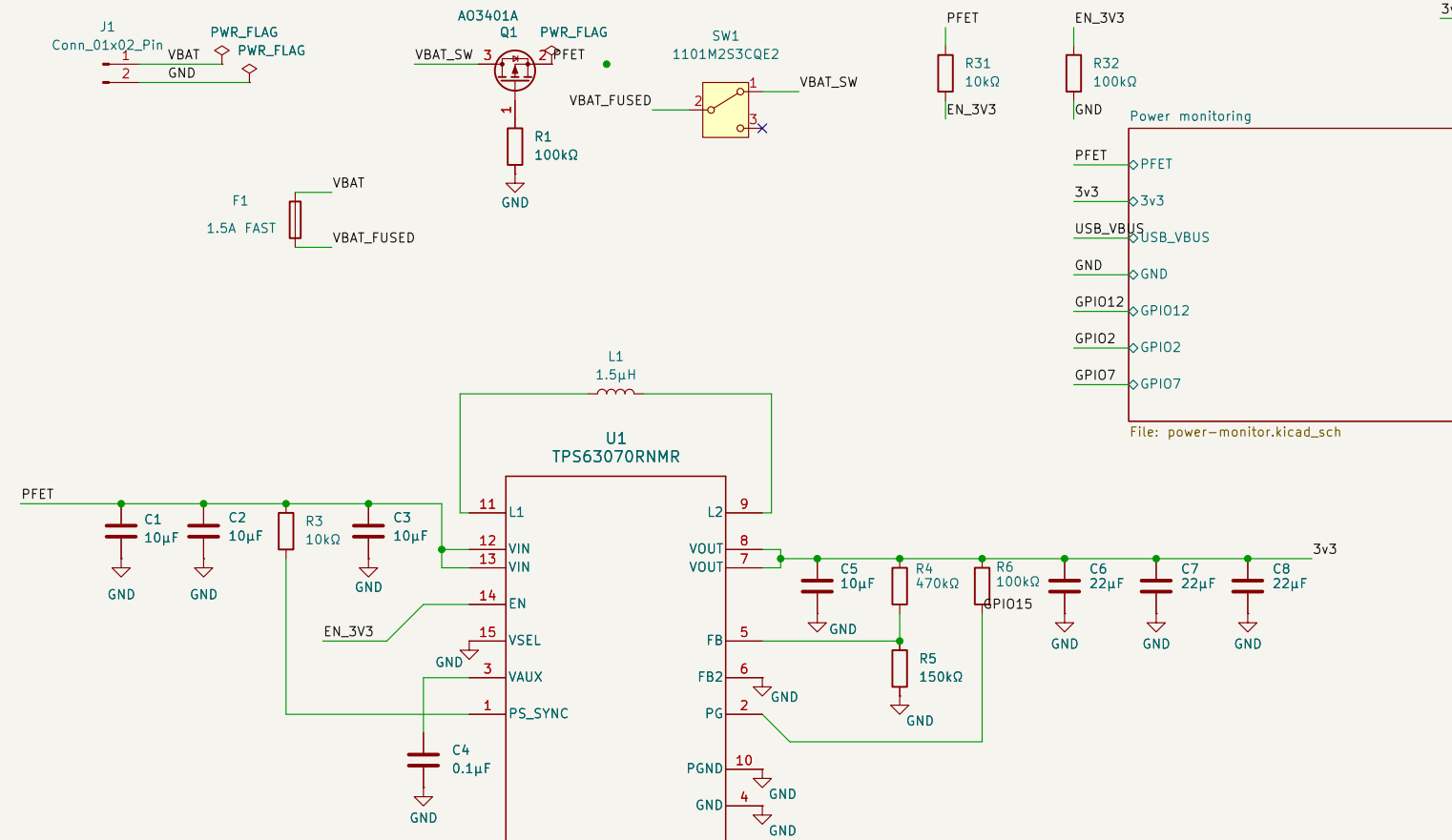
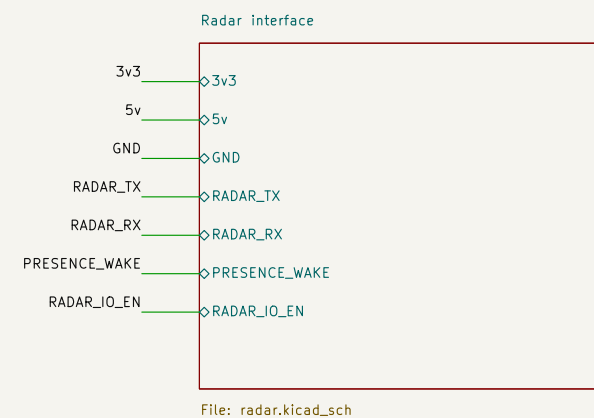
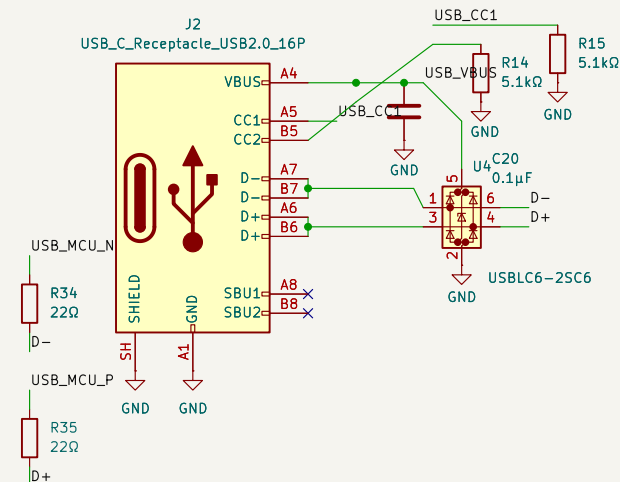
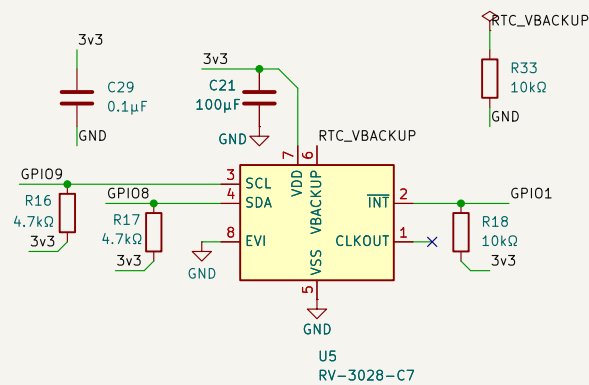
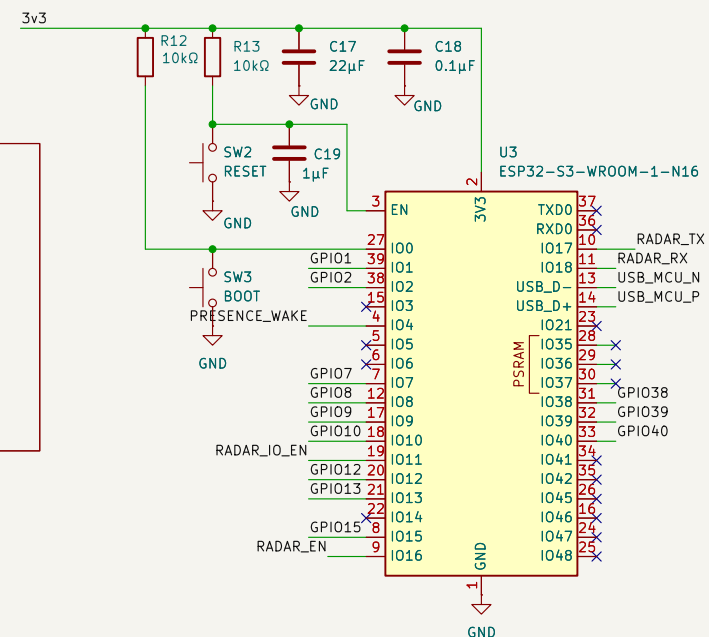


ALEC SENSOR S1 – prototype; bench and ingress tests required.
3 AA primary cells; no charging. USB-C is data-only; batteries + SW1 ON required.
SW1: hard battery disconnect, 6 A / 28 VDC. F1: 1.5 A fast fuse; validate inrush and fault clearing.
RTC time survives ESP deep sleep, but is lost on SW1 OFF or battery removal. Resync before arming.

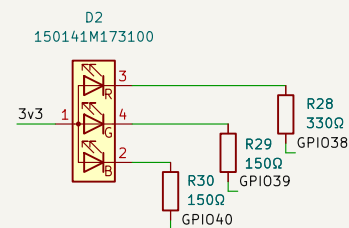
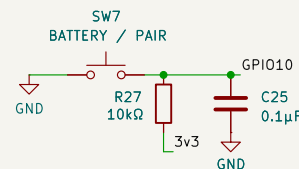


3.31 V always on with SW1 closed. 4.985 V radar rail: GPIO16 high = ON.
R19 holds radar OFF during reset; PS/SYNC high permits power-save operation.
Reserve >=200 mA at the radar connector; measure startup at depleted-cell voltage.

```
GPIO17 TX -> radar RX; GPIO18 RX <- radar TX. UART: 256000 8N1.
GPIO4 = presence wake. GPIO11 = I/O switch enable, default OFF.
Power up: I/O OFF -> rail ON -> PG valid -> I/O ON -> fresh valid frames.
Power down: I/O OFF -> UART stop -> rail OFF. OUT cannot wake if radar is off.
```



RV-3028: I2C GPIO8 SDA / GPIO9 SCL; interrupt GPIO1 (active LOW).
Short press GPIO10: battery; long press: physical pairing.
RGB common anode; GPIO38/39/40 LOW lights R/G/B.
No always-on power LED. Use low-duty PWM for presence indication.

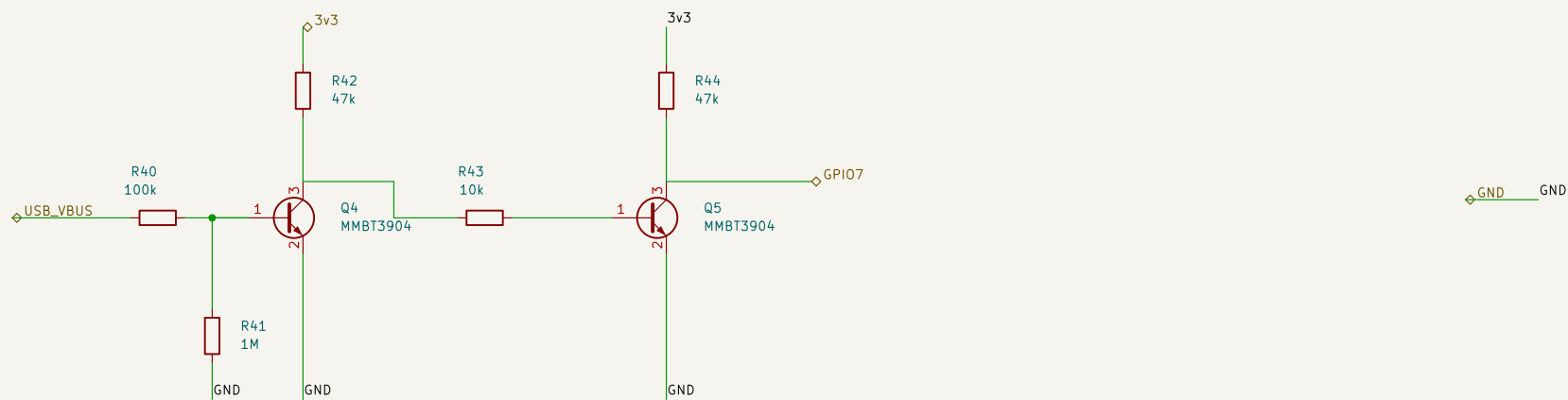


Prototype hardware. See TEST-REPORT.md for checks and remaining bench, radome, condensation and water-ingress release gates.

Sheet: /		
File: alec-sensor.kicad_sch		
Title: ALEC Sensor – prototype S1		
Size: A3	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 1/3

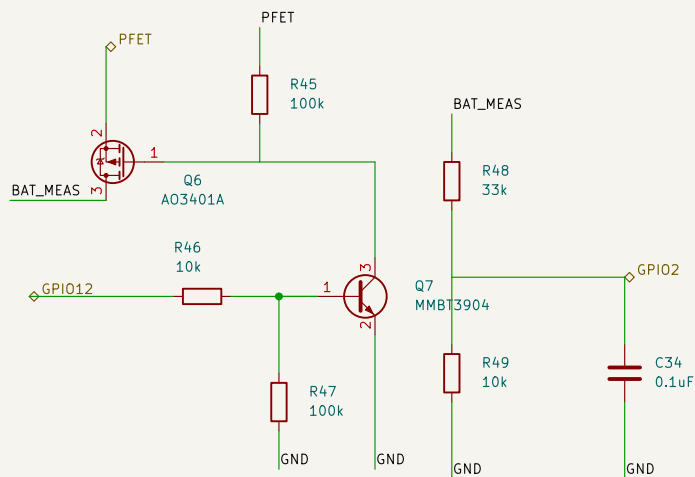
USB VBUS presence – HIGH only with a host supply connected

Two NPN stages keep VBUS away from the GPIO and preserve positive logic.
GPIO7: TinyUSB self_powered=true, vbus_monitor_io=7.
This is cable presence detection, not a precision USB undervoltage comparator.



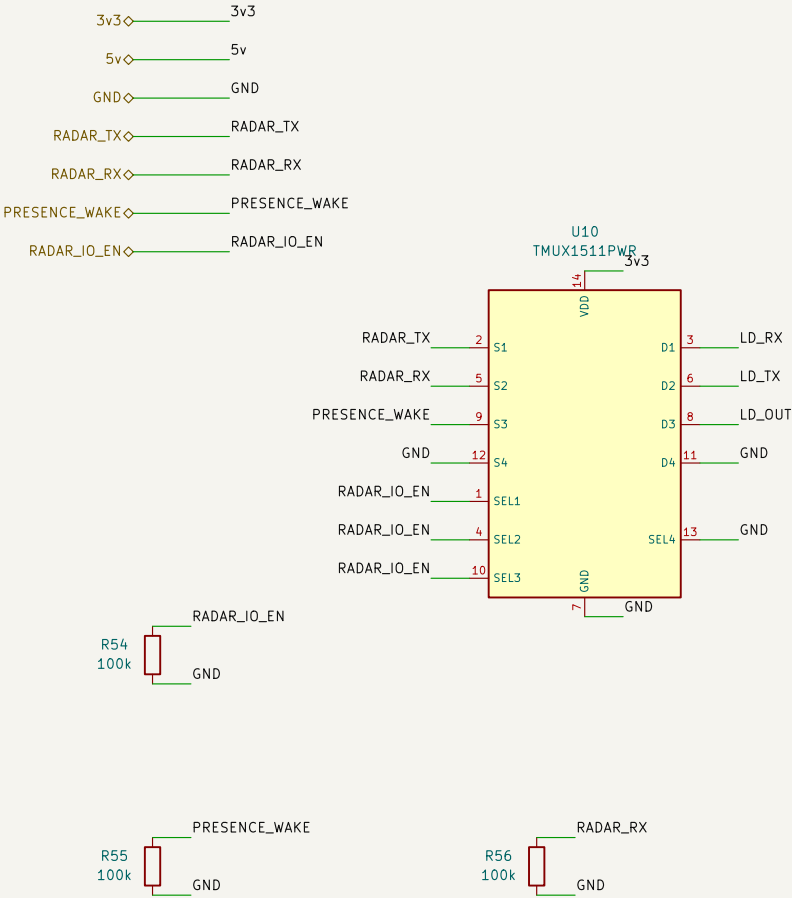
Switched battery measurement – protected battery node to ADC1 / GPIO2

GPIO12 HIGH enables measurement; wait 10 ms, read calibrated ADC, then drive LOW.
Scale = 4.3; 33k / 10k, 0.1%; 100 nF filter, time constant 0.77 ms.
Designed measurement range 0-6 V; battery chemistry and percentage thresholds remain firmware settings.



Sheet: /Power monitoring/		
File: power-monitor.kicad_sch		
Title: ALEC Sensor – power monitoring		
Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 2/3

LD2410C: regulated 5 V; logic is 3.3 V. J3 numbering follows the module drawing.
This is NOT the pin order of a generic JST cable. Verify assembled socket orientation.
U10 opens all three signal paths when GPIO11 is low; no pull-ups on the unpowered radar side.



ON: rail PG valid before closing U10; ignore startup frames until qualified.
OFF: open U10 before disabling 5 V. Check reset, brownout and sleep-hold behavior on the bench.
Powered-radar wake mode: retain 5 V and U10 ON, use OUT only as a wake hint, then verify UART.
U10 is signal isolation, not a voltage translator. The fourth channel is grounded and disabled.

Sheet: /Radar interface/		
File: radar.kicad_sch		
Title: ALEC Sensor – switched radar interface		
Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 3/3